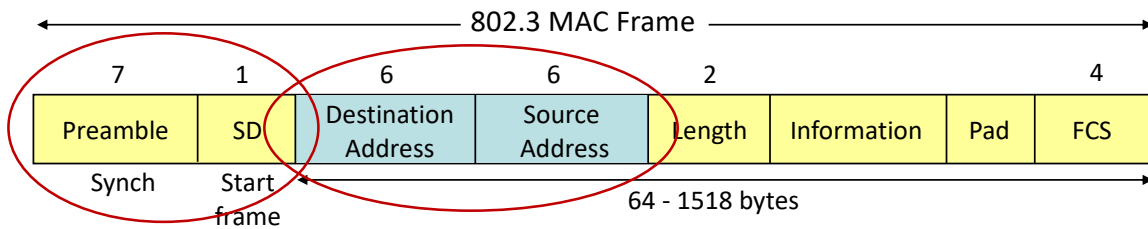


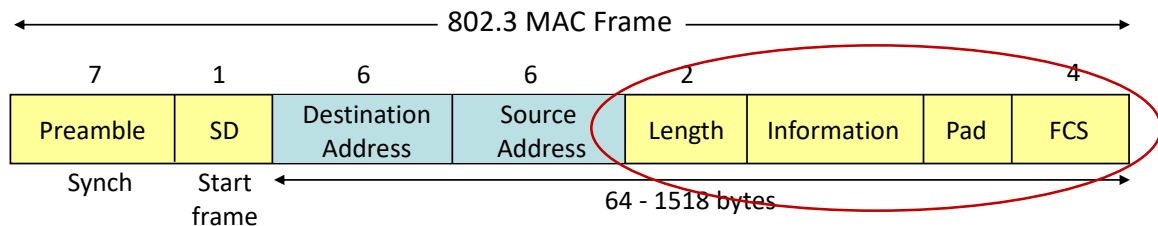
IEEE 802.3 MAC Frame



- ⊕ Preamble helps receiver synchronize its clock to transmitter clock
 - ➡ 7 bytes of 10101010 generate a square wave
- ⊕ Start frame byte changes to “1010101**1**”
- ⊕ Receiver looks for change in “10” pattern
- ⊕ MAC Addresses: 48 bits

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IEEE 802.3 MAC Frame



- ⊕ Length: # of bytes in the information field
 - ➡ Max frame 1518 bytes, excluding preamble & SD
 - ➡ Max information 1500 bytes: 05DC
- ⊕ Pad: ensures **minimum frame size of 64 bytes**
- ⊕ FCS: CRC-32, covers addresses, length, information, and pad fields
 - ➡ NIC discards frames with improper lengths or failed CRC

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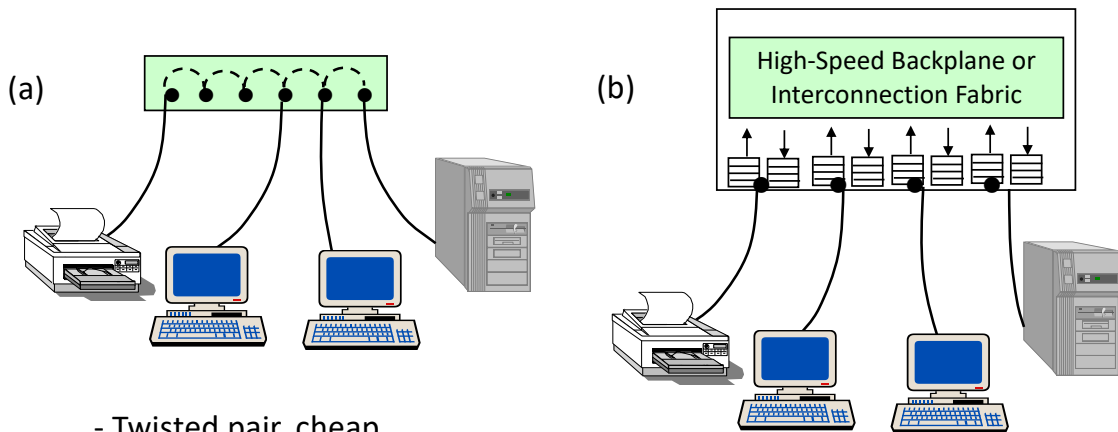
Ethernet Physical Layers

Hubs & Switches

	10base <u>5</u>	10base <u>2</u>	10base <u>T</u>	100base <u>T</u>
Medium	thick coax	thin coax	<u>t</u> wisted pair	CAT5
Max Segment Length	<u>5</u> 00 m	<u>2</u> 00 m	100 m	100 m
Topology	bus	bus	star	star

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Ethernet Hubs & Switches



- Twisted pair, cheap
- Easy to work with
- Star-topology CSMA/CD
- Same collision domain
- Same broadcast domain

- Separate collision domains
- Same broadcast domain

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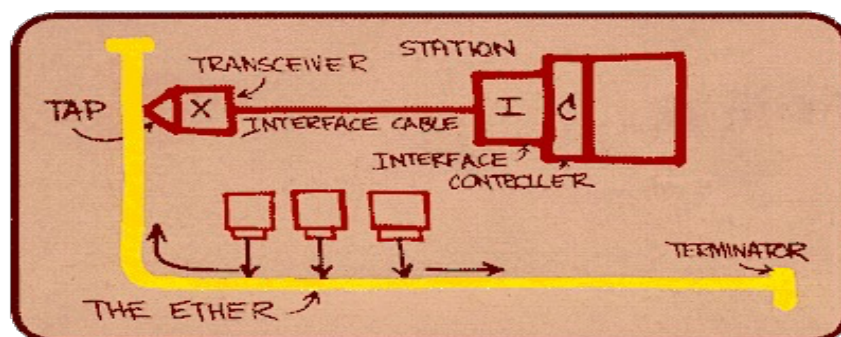
Repeaters, Bridges, Routers, Gateways

- ⊕ Several ways of interconnecting networks:
 - ➡ Repeater at the physical layer
 - ➡ Switch (or bridge) at the MAC or data link layer
 - ➡ Router at the network layer
 - ➡ Gateway at a higher layer

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Ethernet LAN Evolution

- ⊕ 1970 ALOHAnet radio network deployed in Hawaiian islands
- ⊕ 1973 Metcalf and Boggs invented Ethernet: *random access in wired net*
- ⊕ 1985 IEEE 802.3 LAN Standard (10 Mbps)
- ⊕ 1995 Fast Ethernet (100 Mbps)
- ⊕ 1998 Gigabit Ethernet
- ⊕ 2002 10 Gigabit Ethernet
- ⊕ 2007 100 Gigabit Ethernet
- ⊕ Ethernet is the dominant LAN standard



Metcalf's Famous Sketch

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